

# Manav Rath

Software Engineer with 4 years and 6 months of experience specializing in cloud infrastructure, DevOps, and backend development. Skilled in Python, Java, Kubernetes, and CI/CD automation, with hands-on expertise in GCP, AWS, and Azure. Passionate about building scalable and efficient solutions.

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## EXPERIENCE

|                       |                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| Emerging Technologies | Centric8, AI (LLM Model Training), Containerisation using Docker, Kubernetes                           |
| Operating System      | Red Hat Enterprise Linux, CentOS, Ubuntu, Rocky                                                        |
| Platform              | GCP, Azure, Heroku, AWS                                                                                |
| CI/CD                 | Github Actions, Workflows, Shell Scripting                                                             |
| Database              | MSSQL, Redis, PostgreSQL, ElasticSearch, Cassandra, Snowflake                                          |
| AI                    | Kiro, Claude, CoPilot, Gemini, NotebookLM, Deepseek, ChatGPT, Prompt Writing, Model Deployment, Tuning |
| Automation            | Terraform, Ansible, Jenkins                                                                            |
| Networking            | TCP/IP, DNS, NFS, SSL, SAML, OAUTH2                                                                    |
| High Availability     | GCP, Azure, Docker, Containerd, Kubernetes, OpenShift                                                  |
| Storage               | S3, NAS, LVM                                                                                           |
| Tools                 | Jira, Confluence, Git                                                                                  |
| Monitoring            | ELK Stack, Splunk, Prometheus, Grafana                                                                 |
| Programming           | Python, Java, Shell Scripting, Javascript, C#                                                          |
| Security              | OAuth, RBAC, ABAC, SSO, MFA, 2FA                                                                       |
| Framework             | React, Springboot, Flask, FastAPI, D3js, Next.js                                                       |
| Testing               | Specflow, PyTest, Gherkin, Unittest, junit, NUnit                                                      |
| Logging               | Proxy Logs, DNS Queries, System Health Logging                                                         |
| On Hands Extra Tool   | Configura CET Designer Tool, CM Language Programing, AutoCAD, Altova                                   |

## Known Industries

Aviation, Fashion, Data, Advertisement

## ACCREDITATIONS

Google Kubernetes Certification

Google Cloud Platform  
DevSecOps

Python and Java Trainings

Google Cloud Platform  
Networking

Linux administration

Kubernetes

## INTEREST

Billiards and Pool

Played Several Tournaments at positional level.

Hiking

Running

## LANGUAGES

Fluent in English and Hindi

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| Training |  |
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## Hands On Training

2 Months Hands-on training on CM Language Usage and Debugging using emacs IDE. Configured CET Platform specification and product overview. Implemented Solution in regards to Material Handling Module in time period of Practice as a practitioner.

4 Months Hands-on training on ELK Stack. From platform usage to different functionalities. Learned about indexing and deployment of a full stack ELK System.

2 Years of Hands-on Experience of Centric8 while working with H&M for PLM integration. Learned about the components, attributes per model, adapted to using the Centric8 platform along including API usage.

## EXPERIENCE

### Affle, Gurgaon — *SDET - II (Full Time)*

April 2025 - PRESENT

Affle 3i is a global technology company with a consumer intelligence platform that delivers engagement, acquisitions & transactions through mobile advertising. Affle includes several products, In which I am working with **jampp** so far. It is a campaign manager tool and insights towards the campaigns. My role is to improve the code readability, scalability and coverage and refactoring the code. I will be learning how a system manages billions of requests without hinging.

### Bosler, France — *Technical Expert (Contract)*

March 2024 - April 2025

Bosler is a data analysis platform where I develop and maintain high-performance applications. Developed big data processing pipelines using Java, Python, Hadoop, and Spark, improving data retrieval speed by 30%. Optimized PostgreSQL queries, reducing report generation time from 10 minutes to 2 minutes. Led the Linux server operations, automating system monitoring and log analysis using Bash scripting & ELK Stack. Collaborated with cross - functional teams to design and implement scalable software solutions.

### H&M, Sweden — *Python Developer & GCP Cloud Engineer (Contract)*

March 2022 - March 2024

Led the PLM (Product Lifecycle Management) Architecture setup using Centric8 and RabbitMQ while working on software development and cloud infrastructure. Designed and implemented Python-based automation scripts to streamline CI/CD workflows, Managed proxy configurations & API gateways, ensuring secure and efficient data

flow. Built SQL-based analytics pipelines, enabling real-time insights for key stakeholders.

### ***Skythread Airbus, California — C# Software Engineer (Contract)***

February 2021 - February 2022

Developed and maintained C#-based distributed applications for secure data sharing on Hyperledger Fabric. Designed Azure DevOps CI/CD pipelines, reducing deployment time by 50%. Led unit testing and code reviews using NUnit & SpecFlow, improving code quality and maintainability. Integrated blockchain-based smart contracts with existing enterprise systems.

## **PROJECTS**

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### ***1. Esentri - LLM Training Assignment***

Developed a Python application using dependencies like unSloth and ollama to train a LLaMA 3B model for categorizing emails based on sentiment and subject. This model populated JSON templates, automating email routing to the appropriate departments, improving internal workflows, and fostering better client interaction across the organization.

### ***2. TRS Implementation - OCR Service***

Built an OCR service in Python using pytesseract, scikit-learn, numpy, pandas, and FastAPI to extract text from images and PDFs. The system processes documents and outputs actionable insights related to defects and complaints, providing better visibility and understanding for teams to address issues effectively.

### ***3. Custom CLI - Seamlessly Manage Cloud/On-Premise Services***

Developed a Python-based command-line interface (CLI) that integrates with Heroku, GCP, Azure, AWS, and on-premise services like Kubernetes, Container Registries, Cloud Computing, Storages, Databases, and CI/CD. The tool automates cloud resource management, making it easier for organizations to interact with multiple platforms from a single interface.

### ***4. Heroku Metrics Service - Monitoring & Alerting for Collective Pipelines***

Created a Python application to monitor Heroku's platform, specifically focusing on all teams and pipelines to track access rights and system metrics. The service enables non-technical team members to efficiently observe application performance and shut down redundant services with minimal effort, enhancing operational efficiency across teams.

## ***5. Snap - Platform Version Management System***

Developed a Java Spring Boot application to build Docker images using buildah, optimizing performance during builds. The service integrates with a self-hosted registry (Go Harbor) and manages platform licenses and subscriptions. The system is setup using Terraform, deploying on cloud services like Heroku, GCP, Azure, AWS, as well as on-premise infrastructure.

## ***6. On-Premise Kubernetes Deployment with Helm***

Built and deployed a robust on-premise Kubernetes cluster using Kubeadm, Kubelet, and Kubectl, ensuring scalability and efficient application deployment. Leveraged Helm for package management, supporting hundreds of users and ensuring high availability. This experience deepened my expertise in Kubernetes administration and infrastructure management.

## ***7. End-to-End GKE Infrastructure Automation with Terraform on Google Cloud***

Designed and implemented an end-to-end GKE infrastructure solution using Terraform for automated deployment, configuration, and management of a Google Kubernetes Engine (GKE) cluster on Google Cloud Platform (GCP). Configured highly secure and scalable GKE clusters with preemptible node pools using e2-standard-8 machine types and 50 GB disks. Integrated service accounts and IAM roles for secure access management. Leveraged Terraform modules to optimize the deployment process, resolve SSD storage limitations, and ensure compliance with available quotas. The entire infrastructure was automated, reducing manual intervention, enhancing consistency, and streamlining the provisioning process.

Tools used: Terraform, Google Cloud Platform (GCP), Google Kubernetes Engine (GKE), IAM, Preemptible Node Pools, Service Accounts, Cloud Quotas

## ***8. End-to-End AWS Infrastructure Automation with Terraform***

Designed and implemented an end-to-end AWS infrastructure solution using Terraform for automated deployment, configuration, and management of services across the AWS cloud. Configured highly scalable and secure environments, including Amazon Elastic Kubernetes Service (EKS) clusters with EC2 instances and Elastic Load Balancers (ELBs). Integrated IAM roles, security groups, and VPC configurations to ensure secure access management and network isolation. Leveraged Terraform modules to streamline resource provisioning, optimize infrastructure scalability, and handle resource limitations effectively. Automated infrastructure management processes, reducing manual intervention, enhancing consistency, and enabling quicker provisioning of services. The solution also incorporated AWS-native monitoring tools like CloudWatch for real-time metrics and automated alerts, ensuring continuous system performance and operational efficiency.

Tools used: Terraform, Amazon Web Services (AWS), Elastic Kubernetes Service (EKS), IAM, EC2, VPC, CloudWatch, Elastic Load Balancers (ELBs).

## ***9. End-to-End On-Premise Kubernetes Architecture with Calico and MetalLB***

Designed and implemented a fully automated on-premise Kubernetes architecture using Calico for networking and MetalLB for load balancing. Configured a scalable, highly available Kubernetes cluster with optimized network policies for secure communication between pods. Integrated Calico CNI for fine-grained network policy management and MetalLB for efficient external load balancing, ensuring seamless access to services. Utilized Kubeadm for cluster setup and management, while also employing Helm for package management and deployment automation. The architecture was designed to support multiple applications with high availability, reducing manual intervention and improving operational efficiency.

Tools used: Kubernetes, Calico CNI, MetalLB, Kubeadm, Helm, Containerd, Snap(Self Service)

## ***10. On-Premise NAS for Cloud Storage & Backup Management***

Designed and implemented an on-premise NAS solution to serve as a centralized cloud storage system with automated backups across multiple networks. Integrated RAID management for data redundancy and integrity, offering a scalable, secure, and cost-effective backup system that ensures reliable data storage and recovery.

## ***11. GoHarbor Container Registry Setup***

Configured and managed a private GoHarbor container registry for storing and distributing Docker container images securely. Integrated GoHarbor with CI/CD pipelines to enable automated image builds and deployments across different environments. Implemented security policies to ensure secure storage and management of images. Additionally, set up Helm charts for streamlined application deployments, enabling easy management of containerized workloads across cloud and on-premise environments.

Tools used: GoHarbor, Docker, CI/CD Pipelines, Kubernetes, Helm

## ***12. Automated Fleet-Managed ELK Architecture for Multi-Environment Monitoring with SSL/TLS Encryption***

Designed and implemented an automated fleet-managed ELK stack (Elasticsearch, Logstash, Kibana) architecture to monitor and analyze logs across multiple environments in different countries. Integrated Filebeat, Metricbeat, Packetbeat, Heartbeat, Elastic-Agent, and Fleet for centralized logging, system metrics, APM, and security event monitoring across a diverse infrastructure, ensuring real-time insights into application and system performance. The architecture enabled efficient monitoring of Kubernetes clusters, with automated log collection, aggregation, and visualization for performance analysis and troubleshooting. Configured Elasticsearch for log aggregation, Logstash for data pipeline processing, and Kibana for actionable insights and customizable dashboards.

To ensure security across distributed environments, I automated the setup of SSL certificates using Let's Encrypt, securing communication between components like Kibana, Elasticsearch, and the Beats agents. Implemented Certbot to automate SSL/TLS certificate provisioning and renewal, ensuring encrypted communication and minimizing manual intervention. The automatic certificate lifecycle management ensured continuous encryption for all services, including internal and external communication between the monitoring stack and users.

